TASK 4: TYRE WEAR ROOT-CAUSE AND GEOMETRY INVESTIGATION

Autocare Technician Level 2 | Block 8 | v2.1

TIME	EVIDENCE	PROGRAMME POINT
60 minutes	**Practical | Knowledge | Documentation**	**Weeks 71-80**

Your task

A vehicle has excessive or uneven tyre wear. You must inspect the tyres and related steering/suspension geometry, record evidence and identify the likely root cause before recommending work.

What you must do

1. Prepare the vehicle and complete tyre/alignment pre-checks.
2. Record tyre wear patterns and pressures.
3. Measure or interpret alignment/geometry data as instructed.
4. Identify likely root cause and recommended action.
5. Record findings and reinstate the workstation.

What good evidence looks like

SAFE SETUP
Suitable PPE, a prepared work area and controlled vehicle or equipment.

CORRECT METHOD
A logical sequence using the relevant procedure and technical information.

ACCURATE EVIDENCE
Actual readings, findings and actions recorded clearly and precisely.

SOUND DECISIONS
Appropriate recommendations, referral or authorisation supported by evidence.

You will need

- Light vehicle
- Tread depth gauge and tyre pressure gauge
- Alignment equipment or sample printouts
- Manufacturer geometry data
- PPE and Apprentice Assessment Record

SOW ALIGNMENT

Block 8 - Advanced Diagnostics and Workshop Management
Duties: 1-10 KSB evidence focus: K15-K21, S10-S15, B1, B2
Task reference: Block_8_Task_4_Tyre_Wear_Root_Cause

TASK 4: TYRE WEAR ROOT-CAUSE AND GEOMETRY INVESTIGATION

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

TASK COMPLETED AT

○ **APPRENTICE'S WORKPLACE**

○ **TRAINING CENTRE**

Vehicle make

Model

Registration

Mileage

Engine size / type

VIN / chassis number

Measurements and technical checks

Tyre wear pattern		NS/F tread depth	
OS/F tread depth		Front toe measured	
Front toe specification		Camber measured	
Tyre pressure		Suspected root cause	

Special tools, equipment or calibration requirements

Task evidence

PPE and precautions used

Safety-critical findings and immediate action

Work completed, findings and recommendations

Knowledge check

1. What tyre wear pattern can incorrect toe create?

2. Why must suspension condition be checked before alignment?

Knowledge check - continued

3. (a) How can tyre pressure affect wear patterns? (b) Why is rear alignment or thrust angle important?

4. How would you explain the benefits of correct geometry?

5. (a) What evidence confirms root cause rather than symptom? (b) What recommendation should be made if a component is worn?

Assessor/Workplace Mentor observation

- ☐ Inspect tyre wear patterns and related components.
- ☐ Record geometry or alignment measurements where available.
- ☐ Identify likely root causes using evidence.
- ☐ Prioritise repairs, adjustment or referral.
- ☐ Explain the customer benefit of correct geometry.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- ○ **PASS**
- ○ **REFER**

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

PIN (if applicable)

Apprentice signature

Assessor/Workplace Mentor signature

Date